

CS 550 – PA3
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Output File

A2A Topology Outputs

- Screenshot of 9 super-peers running with all-to-all topology configuration. Steps used to start each peer:
 - cd src
 - ./run_peer 1 A2A //to start server-peer 1 with A2A topology.

```
(base) fatimamariyam@Fatimas-MacBook-Air src % ./run_peer.sh 1 A2A
*****
Super-peer 1
available directory ../dirs/Peer1
Topology: all-to-all-topology.txt
*****
Enter the filename to download

--- Running..... Ready to receive download request ---

[

(base) fatimamariyam@Fatimas-MacBook-Air src % ./run_peer.sh 4 A2A
*****
Super-peer 4
available directory ../dirs/Peer4
Topology: all-to-all-topology.txt
*****
Enter the filename to download

--- Running..... Ready to receive download request ---

[

Last login: Tue Mar 26 13:59:12 on ttys026
(base) fatimamariyam@Fatimas-MacBook-Air src % ./run_peer.sh 7 A2A
*****
Super-peer 7
available directory ../dirs/Peer7
Topology: all-to-all-topology.txt
*****
Enter the filename to download

--- Running..... Ready to receive download request ---

[

(base) fatimamariyam@Fatimas-MacBook-Air src % ./run_peer.sh 2 A2A
*****
Super-peer 2
available directory ../dirs/Peer2
Topology: all-to-all-topology.txt
*****
Enter the filename to download

--- Running..... Ready to receive download request ---

[

(base) fatimamariyam@Fatimas-MacBook-Air src % ./run_peer.sh 5 A2A
*****
Super-peer 5
available directory ../dirs/Peer5
Topology: all-to-all-topology.txt
*****
Enter the filename to download

--- Running..... Ready to receive download request ---

[

Last login: Tue Mar 26 13:59:09 on ttys025
(base) fatimamariyam@Fatimas-MacBook-Air src % ./run_peer.sh 8 A2A
*****
Super-peer 8
available directory ../dirs/Peer8
Topology: all-to-all-topology.txt
*****
Enter the filename to download

--- Running..... Ready to receive download request ---

[

Last login: Tue Mar 26 13:59:15 on ttys027
(base) fatimamariyam@Fatimas-MacBook-Air src % ./run_peer.sh 6 A2A
*****
Super-peer 6
available directory ../dirs/Peer6
Topology: all-to-all-topology.txt
*****
Enter the filename to download

--- Running..... Ready to receive download request ---

[

Last login: Tue Mar 26 13:59:19 on ttys028
(base) fatimamariyam@Fatimas-MacBook-Air src % ./run_peer.sh 9 A2A
*****
Super-peer 9
available directory ../dirs/Peer9
Topology: all-to-all-topology.txt
*****
Enter the filename to download

--- Running..... Ready to receive download request ---

[
```

- Directory structure:

```
-Peer10
|-file(10)_4.txt
|-file(10)_5.txt
|-file(10)_2.txt
|-file(10)_3.txt
|-file(10)_1.txt
-Peer5
|-file(5)_2.txt
|-file(5)_1.txt
-Peer2
|-file(2)_3.txt
|-file(2)_2.txt
|-file(2)_1.txt
|-file(2)_4.txt
-Peer3
|-file(3)_1.txt
|-file(3)_2.txt
|-file(3)_3.txt
|-file(3)_7.txt
|-file(1)_3.txt
|-file(3)_6.txt
|-file(3)_4.txt
|-file(3)_5.txt
|-file(3)_8.txt
-Peer4
|-file(4)_1.txt
|-file(4)_2.txt
|-file(4)_3.txt
|-file(4)_7.txt
|-file(4)_6.txt
|-file(4)_4.txt
|-file(4)_5.txt
-Peer8
|-file(8)_1.txt
|-file(8)_2.txt
|-file(8)_3.txt
|-file(8)_6.txt
|-file(8)_4.txt
|-file(8)_5.txt
-Peer1
|-file(1)_4.txt
```

```

|-file(1)_5.txt
|-file(1)_6.txt
|-file(1)_2.txt
|-file(1)_3.txt
|-file(1)_1.txt
-Peer6
|-file(6)_4.txt
|-file(6)_5.txt
|-file(6)_2.txt
|-file(6)_3.txt
|-file(6)_1.txt
-Peer7
|-file(7)_6.txt
|-file(7)_7.txt
|-file(7)_5.txt
|-file(7)_4.txt
|-file(7)_1.txt
|-file(7)_10.txt
|-file(7)_3.txt
|-file(7)_2.txt
|-file(7)_9.txt
|-file(7)_8.txt
-Peer9
|-file(9)_3.txt
|-file(9)_2.txt
|-file(9)_1.txt
|-file(9)_5.txt
|-file(9)_4.txt

```

- Example: query and download **file(6)_1.txt** from peer1

Server-Peer1 Output

file(6)_1.txt

neighbours --> [2, 3, 4, 5, 6, 7, 8, 9]

```

[
Leaf-Peer {SuperPeer=2, port=4001, filetodownload=file(6)_1.txt, fromPeer=1, message-id=1.1},
Leaf-Peer {SuperPeer=3, port=4002, filetodownload=file(6)_1.txt, fromPeer=1, message-id=1.1},
Leaf-Peer {SuperPeer=4, port=4003, filetodownload=file(6)_1.txt, fromPeer=1, message-id=1.1},
Leaf-Peer {SuperPeer=5, port=4004, filetodownload=file(6)_1.txt, fromPeer=1, message-id=1.1},
Leaf-Peer {SuperPeer=6, port=4005, filetodownload=file(6)_1.txt, fromPeer=1, message-id=1.1},
Leaf-Peer {SuperPeer=7, port=4006, filetodownload=file(6)_1.txt, fromPeer=1, message-id=1.1},
Leaf-Peer {SuperPeer=8, port=4007, filetodownload=file(6)_1.txt, fromPeer=1, message-id=1.1},
Leaf-Peer {SuperPeer=9, port=4008, filetodownload=file(6)_1.txt, fromPeer=1, message-id=1.1}]

```

leaf-peer containing the file:

6

Selecting leaf-peer: 6 for file download

file(6)_1.txt file is transferred to your private storage: ../dirs/Peer1

File: file(6)_1.txt downloaded from leaf-peer: 6 to leaf-peer: 1

Server-Peer2 Output

Superpeer: 2, started at /127.0.0.1:53001

New Query received from peer with peer-id : 1

Superpeer: "Search in leaf-peer completed"

Query forwarded to next neighbour: 3

Query forwarded to next neighbour: 4

Query forwarded to next neighbour: 5

Query forwarded to next neighbour: 6

Query forwarded to next neighbour: 7

Superpeer: 2, started at /127.0.0.1:53006

Query forwarded to next neighbour: 8

Query forwarded to next neighbour: 9

Superpeer: 2, started at /127.0.0.1:53008

Superpeer: 2, started at /127.0.0.1:53011

New Query received from peer with peer-id : 9

Same query received!

Superpeer: 2, started at /127.0.0.1:53012

New Query received from peer with peer-id : 4
Same query received!

Superpeer: 2, started at /127.0.0.1:53036
New Query received from peer with peer-id : 5
Same query received!

Superpeer: 2, started at /127.0.0.1:53045
New Query received from peer with peer-id : 8
Same query received!

Superpeer: 2, started at /127.0.0.1:53049
New Query received from peer with peer-id : 3
Same query received!
New Query received from peer with peer-id : 6
Same query received!
New Query received from peer with peer-id : 7
Same query received!

Server-Peer3 Output

Superpeer: 3, started at /127.0.0.1:53005
New Query received from peer with peer-id : 1
Superpeer: "Search in leaf-peer completed"
Query forwarded to next neighbour: 2
Query forwarded to next neighbour: 4
Query forwarded to next neighbour: 5
Query forwarded to next neighbour: 6
Query forwarded to next neighbour: 7
Query forwarded to next neighbour: 8

Superpeer: 3, started at /127.0.0.1:53007
Query forwarded to next neighbour: 9

Superpeer: 3, started at /127.0.0.1:53015
New Query received from peer with peer-id : 9
Same query received!

Superpeer: 3, started at /127.0.0.1:53021

Superpeer: 3, started at /127.0.0.1:53022
New Query received from peer with peer-id : 4
Same query received!

Superpeer: 3, started at /127.0.0.1:53042
New Query received from peer with peer-id : 8
Same query received!

Superpeer: 3, started at /127.0.0.1:53048
New Query received from peer with peer-id : 5
Same query received!

Superpeer: 3, started at /127.0.0.1:53054
New Query received from peer with peer-id : 2
Same query received!
New Query received from peer with peer-id : 6
Same query received!
New Query received from peer with peer-id : 7
Same query received!

Super-peer4 Output

Superpeer: 4, started at /127.0.0.1:53000
New Query received from peer with peer-id : 1
Superpeer: "Search in leaf-peer completed"
Query forwarded to next neighbour: 2
Query forwarded to next neighbour: 3
Query forwarded to next neighbour: 5
Query forwarded to next neighbour: 6
Query forwarded to next neighbour: 7
Query forwarded to next neighbour: 8
Query forwarded to next neighbour: 9

Superpeer: 4, started at /127.0.0.1:53010

Superpeer: 4, started at /127.0.0.1:53017

Superpeer: 4, started at /127.0.0.1:53020

Superpeer: 4, started at /127.0.0.1:53037
New Query received from peer with peer-id : 9
Same query received!

Superpeer: 4, started at /127.0.0.1:53047
New Query received from peer with peer-id : 5
Same query received!

Superpeer: 4, started at /127.0.0.1:53051
New Query received from peer with peer-id : 8
Same query received!

Superpeer: 4, started at /127.0.0.1:53057
New Query received from peer with peer-id : 3
Same query received!
New Query received from peer with peer-id : 2
Same query received!
New Query received from peer with peer-id : 6
Same query received!
New Query received from peer with peer-id : 7
Same query received!

Super-peer5 Output

Superpeer: 5, started at /127.0.0.1:53004
New Query received from peer with peer-id : 1
Superpeer: "Search in leaf-peer completed"
Query forwarded to next neighbour: 2
Query forwarded to next neighbour: 3
Query forwarded to next neighbour: 4
Query forwarded to next neighbour: 6
Query forwarded to next neighbour: 7
Query forwarded to next neighbour: 8
Query forwarded to next neighbour: 9

Superpeer: 5, started at /127.0.0.1:53013

Superpeer: 5, started at /127.0.0.1:53019

Superpeer: 5, started at /127.0.0.1:53029

Superpeer: 5, started at /127.0.0.1:53038
New Query received from peer with peer-id : 9
Same query received!
New Query received from peer with peer-id : 4
Same query received!

Superpeer: 5, started at /127.0.0.1:53050
New Query received from peer with peer-id : 8
Same query received!

Superpeer: 5, started at /127.0.0.1:53055
New Query received from peer with peer-id : 3
Same query received!

Superpeer: 5, started at /127.0.0.1:53060
New Query received from peer with peer-id : 2
Same query received!
New Query received from peer with peer-id : 6
Same query received!
New Query received from peer with peer-id : 7
Same query received!

Super-peer 6 Output

Superpeer: 6, started at /127.0.0.1:53002
New Query received from peer with peer-id : 1
File Found! queryhit: file(6)_1.txt
Superpeer: "Search in leaf-peer completed"
Query forwarded to next neighbour: 2
Query forwarded to next neighbour: 3
Query forwarded to next neighbour: 4
Query forwarded to next neighbour: 5
Query forwarded to next neighbour: 7
Query forwarded to next neighbour: 8

Superpeer: 6, started at /127.0.0.1:53009
Query forwarded to next neighbour: 9

Superpeer: 6, started at /127.0.0.1:53024

Superpeer: 6, started at /127.0.0.1:53028
New Query received from peer with peer-id : 9
Same query received!

Superpeer: 6, started at /127.0.0.1:53032

Superpeer: 6, started at /127.0.0.1:53039
New Query received from peer with peer-id : 4
Same query received!

Superpeer: 6, started at /127.0.0.1:53041
New Query received from peer with peer-id : 5
Same query received!
New Query received from peer with peer-id : 8
Same query received!

Superpeer: 6, started at /127.0.0.1:53053
New Query received from peer with peer-id : 3

Same query received!
New Query received from peer with peer-id : 7
Same query received!
New Query received from peer with peer-id : 2
Same query received!

Super-peer 7 Output

Superpeer: 7, started at /127.0.0.1:52998
New Query received from peer with peer-id : 1
Superpeer: "Search in leaf-peer completed"
Query forwarded to next neighbour: 2
Query forwarded to next neighbour: 3
Query forwarded to next neighbour: 4
Query forwarded to next neighbour: 5
Query forwarded to next neighbour: 6

Superpeer: 7, started at /127.0.0.1:53023
Query forwarded to next neighbour: 8
Query forwarded to next neighbour: 9

Superpeer: 7, started at /127.0.0.1:53025

Superpeer: 7, started at /127.0.0.1:53030
New Query received from peer with peer-id : 8
Same query received!

Superpeer: 7, started at /127.0.0.1:53031
New Query received from peer with peer-id : 9
Same query received!

Superpeer: 7, started at /127.0.0.1:53035
New Query received from peer with peer-id : 4
Same query received!

Superpeer: 7, started at /127.0.0.1:53058
New Query received from peer with peer-id : 5
Same query received!

Superpeer: 7, started at /127.0.0.1:53059
New Query received from peer with peer-id : 3
Same query received!
New Query received from peer with peer-id : 2
Same query received!
New Query received from peer with peer-id : 6
Same query received!

Super-peer 8 Output

Superpeer: 8, started at /127.0.0.1:52999
New Query received from peer with peer-id : 1
Superpeer: "Search in leaf-peer completed"
Query forwarded to next neighbour: 2
Query forwarded to next neighbour: 3
Query forwarded to next neighbour: 4
Query forwarded to next neighbour: 5
Query forwarded to next neighbour: 6
Query forwarded to next neighbour: 7
Query forwarded to next neighbour: 9

Superpeer: 8, started at /127.0.0.1:53016

Superpeer: 8, started at /127.0.0.1:53018

Superpeer: 8, started at /127.0.0.1:53033

Superpeer: 8, started at /127.0.0.1:53034

Superpeer: 8, started at /127.0.0.1:53044

New Query received from peer with peer-id : 5

Same query received!

New Query received from peer with peer-id : 9

Same query received!

New Query received from peer with peer-id : 4

Same query received!

Superpeer: 8, started at /127.0.0.1:53046

Superpeer: 8, started at /127.0.0.1:53061

New Query received from peer with peer-id : 7

Same query received!

New Query received from peer with peer-id : 3

Same query received!

New Query received from peer with peer-id : 6

Same query received!

New Query received from peer with peer-id : 2

Same query received!

Super-peer 9 Output

Superpeer: 9, started at /127.0.0.1:53003

New Query received from peer with peer-id : 1

Superpeer: "Search in leaf-peer completed"

Query forwarded to next neighbour: 2

Query forwarded to next neighbour: 3

Query forwarded to next neighbour: 4

Query forwarded to next neighbour: 5

Query forwarded to next neighbour: 6

Query forwarded to next neighbour: 7

Query forwarded to next neighbour: 8

Superpeer: 9, started at /127.0.0.1:53014

Superpeer: 9, started at /127.0.0.1:53026

Superpeer: 9, started at /127.0.0.1:53027

Superpeer: 9, started at /127.0.0.1:53040

New Query received from peer with peer-id : 4

Same query received!

Superpeer: 9, started at /127.0.0.1:53043

New Query received from peer with peer-id : 8

Same query received!

New Query received from peer with peer-id : 5

Same query received!

Superpeer: 9, started at /127.0.0.1:53052

New Query received from peer with peer-id : 3

Same query received!

Superpeer: 9, started at /127.0.0.1:53056

New Query received from peer with peer-id : 2

Same query received!

New Query received from peer with peer-id : 7

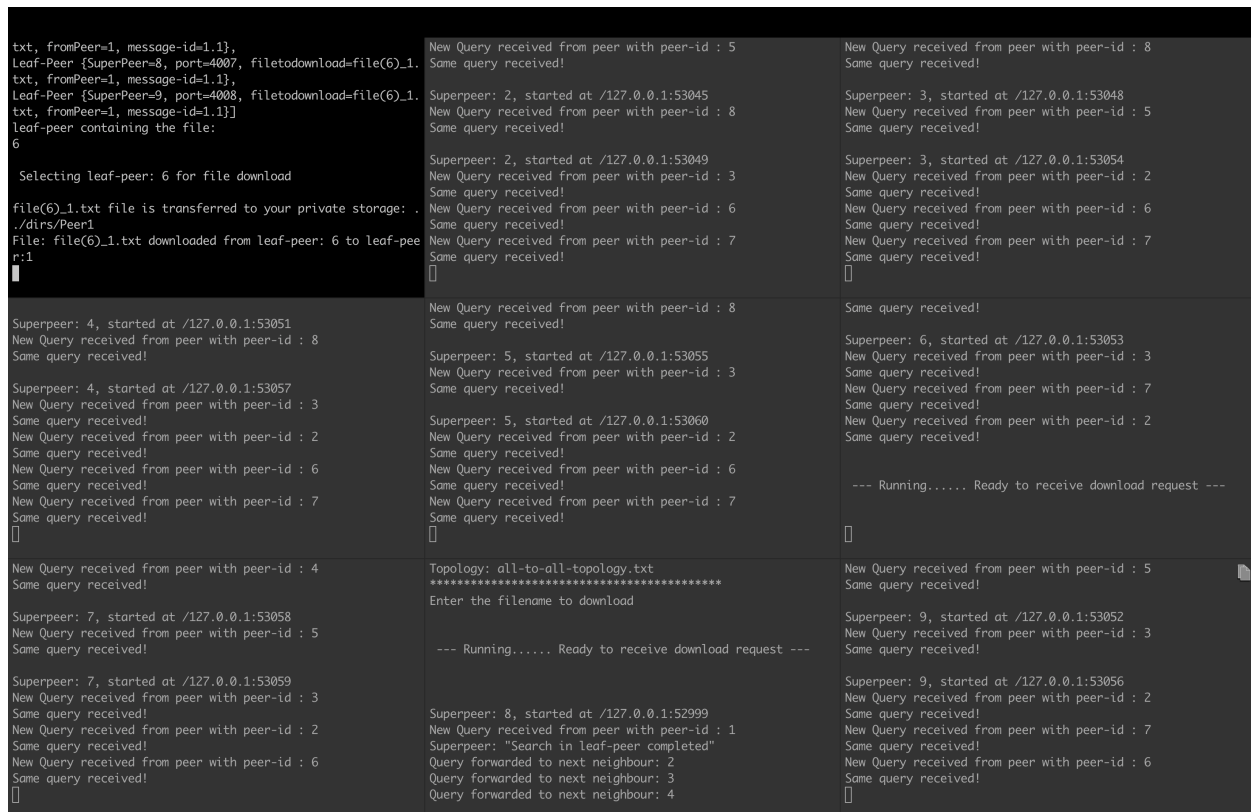
Same query received!

New Query received from peer with peer-id : 6

Same query received!

- We can now see the file file(6)_1.txt downloaded and present in the Peer1 directory.

```
(base) fatimamariyam@Fatimas-MacBook-Air Peer1 % find . | sed -e "s/[^\-][^\-]*\// | /g" -e "s/|\\([^\- ]\\)|-\\1/"
.
|-file(1)_4.txt
|-file(1)_5.txt
|-file(1)_6.txt
|-file(1)_2.txt
|-file(1)_3.txt
|-file(1)_1.txt
|-file(6)_1.txt
```



Tree Topology Outputs

- Screenshot of 9 super-peers running with tree topology configuration. Steps used to start each peer:
 - cd src
 - ./run_peer 1 T //to start server-peer 1 with tree topology.

<pre>(base) fatimamariyam@Fatimas-MacBook-Air src % ./run_peer.sh 1 T ***** Super-peer 1 available directory ../dirs/Peer1 Topology: tree-topology.txt ***** Enter the filename to download --- Running..... Ready to receive download request --- []</pre>	<pre>(base) fatimamariyam@Fatimas-MacBook-Air src % ./run_peer.sh 2 T ***** Super-peer 2 available directory ../dirs/Peer2 Topology: tree-topology.txt ***** Enter the filename to download --- Running..... Ready to receive download request --- []</pre>	<pre>(base) fatimamariyam@Fatimas-MacBook-Air src % ./run_peer.sh 3 T ***** Super-peer 3 available directory ../dirs/Peer3 Topology: tree-topology.txt ***** Enter the filename to download --- Running..... Ready to receive download request --- []</pre>
<pre>(base) fatimamariyam@Fatimas-MacBook-Air src % ./run_peer.sh 4 T ***** Super-peer 4 available directory ../dirs/Peer4 Topology: tree-topology.txt ***** Enter the filename to download --- Running..... Ready to receive download request --- []</pre>	<pre>(base) fatimamariyam@Fatimas-MacBook-Air src % ./run_peer.sh 5 T ***** Super-peer 5 available directory ../dirs/Peer5 Topology: tree-topology.txt ***** Enter the filename to download --- Running..... Ready to receive download request --- []</pre>	<pre>(base) fatimamariyam@Fatimas-MacBook-Air src % ./run_peer.sh 6 T ***** Super-peer 6 available directory ../dirs/Peer6 Topology: tree-topology.txt ***** Enter the filename to download --- Running..... Ready to receive download request --- []</pre>
<pre>(base) fatimamariyam@Fatimas-MacBook-Air src % ./run_peer.sh 7 T ***** Super-peer 7 available directory ../dirs/Peer7 Topology: tree-topology.txt ***** Enter the filename to download --- Running..... Ready to receive download request --- []</pre>	<pre>(base) fatimamariyam@Fatimas-MacBook-Air src % ./run_peer.sh 8 T ***** Super-peer 8 available directory ../dirs/Peer8 Topology: tree-topology.txt ***** Enter the filename to download --- Running..... Ready to receive download request --- []</pre>	<pre>(base) fatimamariyam@Fatimas-MacBook-Air src % ./run_peer.sh 9 T ***** Super-peer 9 available directory ../dirs/Peer9 Topology: tree-topology.txt ***** Enter the filename to download --- Running..... Ready to receive download request --- []</pre>

- Example: query and download **file(6)_1.txt** from peer1

Server-Peer1 Output

file(6)_1.txt

neighbours --> [2, 3]

```
[
Leaf-Peer {SuperPeer=2, port=4001, filetodownload=file(6)_1.txt, fromPeer=1, message-id=1.1},
Leaf-Peer {SuperPeer=3, port=4002, filetodownload=file(6)_1.txt, fromPeer=1, message-id=1.1}]
leaf-peer containing the file:
6
```

Selecting leaf-peer: 6 for file download

file(6)_1.txt file is transferred to your private storage: ../dirs/Peer1

File: file(6)_1.txt downloaded from leaf-peer: 6 to leaf-peer:1

Server-Peer2 Output

```
Superpeer: 2, started at /127.0.0.1:53277
New Query received from peer with peer-id : 1
Superpeer: "Search in leaf-peer completed"
Query forwarded to next neighbour: 4
Query forwarded to next neighbour: 5
```

Server-Peer3 Output

```
Superpeer: 3, started at /127.0.0.1:53276
New Query received from peer with peer-id : 1
Superpeer: "Search in leaf-peer completed"
Query forwarded to next neighbour: 6
Query forwarded to next neighbour: 7
```

Server-Peer4 Output

```
Superpeer: 4, started at /127.0.0.1:53280
```

New Query received from peer with peer-id : 2
Superpeer: "Search in leaf-peer completed"
Query forwarded to next neighbour: 8

Server-Peer5 Output

Superpeer: 5, started at /127.0.0.1:53281
New Query received from peer with peer-id : 2
Superpeer: "Search in leaf-peer completed"
Query forwarded to next neighbour: 9

Server-Peer6 Output

Superpeer: 6, started at /127.0.0.1:53279
New Query received from peer with peer-id : 3
File Found! queryhit: file(6)_1.txt
Superpeer: "Search in leaf-peer completed"

Server-Peer7 Output

Superpeer: 7, started at /127.0.0.1:53278
New Query received from peer with peer-id : 3
Superpeer: "Search in leaf-peer completed"

Server-Peer8 Output

Superpeer: 8, started at /127.0.0.1:53283
New Query received from peer with peer-id : 4
Superpeer: "Search in leaf-peer completed"

Server-Peer9 Output

Superpeer: 9, started at /127.0.0.1:53282
New Query received from peer with peer-id : 5
Superpeer: "Search in leaf-peer completed"

- We can now see the file file(6)_1.txt downloaded and present in the Peer1 directory.

```
(base) fatimamariyam@Fatimas-MacBook-Air Peer1 % find . | sed -e "s/[^\-][^\-]*\\// | /g" -e "s/|\\([^\- ]\\)/|-\\1/"
.
|-file(1)_4.txt
|-file(1)_5.txt
|-file(1)_6.txt
|-file(1)_2.txt
|-file(1)_3.txt
|-file(1)_1.txt
|-file(6)_1.txt
```

<pre> available directory ../dirs/Peer1 Topology: tree-topology.txt ***** Enter the filename to download --- Running..... Ready to receive download request --- file(6).1.txt neighbours --> [2, 3] [Leaf-Peer {SuperPeer=2, port=4001, filetodownload=file(6).1.txt, fromPeer=1, message-id=1.1}, Leaf-Peer {SuperPeer=3, port=4002, filetodownload=file(6).1. </pre>	<pre> Topology: tree-topology.txt ***** Enter the filename to download --- Running..... Ready to receive download request --- Superpeer: 2, started at /127.0.0.1:53277 New Query received from peer with peer-id : 1 Superpeer: "Search in leaf-peer completed" Query forwarded to next neighbour: 4 Query forwarded to next neighbour: 5 [] </pre>	<pre> Topology: tree-topology.txt ***** Enter the filename to download --- Running..... Ready to receive download request --- Superpeer: 3, started at /127.0.0.1:53276 New Query received from peer with peer-id : 1 Superpeer: "Search in leaf-peer completed" Query forwarded to next neighbour: 6 Query forwarded to next neighbour: 7 [] </pre>
<pre> available directory ../dirs/Peer4 Topology: tree-topology.txt ***** Enter the filename to download --- Running..... Ready to receive download request --- Superpeer: 4, started at /127.0.0.1:53280 New Query received from peer with peer-id : 2 Superpeer: "Search in leaf-peer completed" Query forwarded to next neighbour: 8 [] </pre>	<pre> available directory ../dirs/Peer5 Topology: tree-topology.txt ***** Enter the filename to download --- Running..... Ready to receive download request --- Superpeer: 5, started at /127.0.0.1:53281 New Query received from peer with peer-id : 2 Superpeer: "Search in leaf-peer completed" Query forwarded to next neighbour: 9 [] </pre>	<pre> --- Running..... Ready to receive download request --- Superpeer: 6, started at /127.0.0.1:53279 New Query received from peer with peer-id : 3 File Found! queryhit: file(6).1.txt Superpeer: "Search in leaf-peer completed" --- Running..... Ready to receive download request --- [] </pre>
<pre> Super-peer 7 available directory ../dirs/Peer7 Topology: tree-topology.txt ***** Enter the filename to download --- Running..... Ready to receive download request --- Superpeer: 7, started at /127.0.0.1:53278 New Query received from peer with peer-id : 3 Superpeer: "Search in leaf-peer completed" [] </pre>	<pre> Super-peer 8 available directory ../dirs/Peer8 Topology: tree-topology.txt ***** Enter the filename to download --- Running..... Ready to receive download request --- Superpeer: 8, started at /127.0.0.1:53283 New Query received from peer with peer-id : 4 Superpeer: "Search in leaf-peer completed" [] </pre>	<pre> Super-peer 9 available directory ../dirs/Peer9 Topology: tree-topology.txt ***** Enter the filename to download --- Running..... Ready to receive download request --- Superpeer: 9, started at /127.0.0.1:53282 New Query received from peer with peer-id : 5 Superpeer: "Search in leaf-peer completed" [] </pre>